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Application No.:

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PATENT PRODUCT/PROCESS

1. IP support service required (Tick the appropriate service)

- ✓ Patent :
B) Trademark :
✓ Copyright :
✓ Designs :

2. Point of contact (PoC)

a) 1st Point of Contact

Name : Mrs. Ranjitha . P
Mobile No. for contact 9442108819
WhatsApp No. 9442108819
Email : ranjitha18993@gmail.com

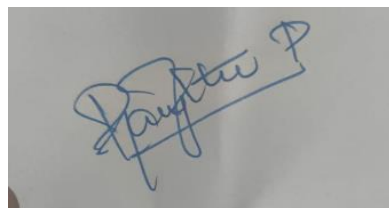
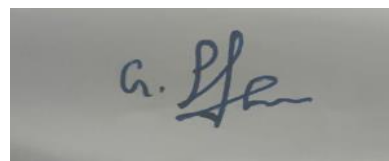
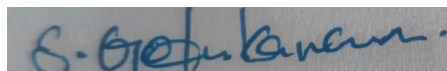
b) 2nd Point of Contact

Name :G.Jagadeesh
Mobile No. for contact 8015069923
WhatsApp No. 8015069923
Email : jaga26304@gmail.com

PATENT PARTICULARS

Name of the Applicant:Mrs.Ranjitha.P

Inventors:

S.No	Inventor Particulars	Signature (Please Attach the JPEG of your signature)
1	Name: Mrs. Ranjitha . P Designation: Assistant Professor Department: ECE Name of the College: MVIT Email: ranjitha18993@gmail.com Mobile No. 9442108819 WhatsApp No. 9442108819	
2	Name: G.Jagadeesh Designation: student Department: ECE Name of the College: MVIT Email: jagadeeshece2023@mvit.edu.in Mobile No. 8015069923 Whatsapp No. 8015069923	
3	Name: M.Bathreesh Designation: student Department: ECE Name of the College: MVIT Email: bathreeshece2023@mvit.edu.in Mobile No. 9087244225 Whatsapp No. 9087244225	
4	Name:S.Gokulakannan Designation: student Department: ECE Name of the College: MVIT Email:gokulakannan2023@mvit.edu.in Mobile No. 9342337945 Whatsapp No. 9342337945	

A. INVENTION TITLE AND SHORT DESCRIPTION

Real-Time FPGA-Accelerated Object Detection and Counting System using PYNQ-Z2

This invention presents a hardware-software co-designed embedded vision system capable of performing real-time object detection using the PYNQ-Z2 FPGA development board. The system integrates the ARM Processing System (PS) and FPGA Programmable Logic (PL) of the Xilinx Zynq-7000 SoC through AXI DMA communication to enable efficient real-time video processing.

The proposed system captures live video frames from a USB camera, transfers image data between PS and PL using AXI DMA, and performs object detection using the YOLOv3-tiny deep learning model running on the ARM Cortex-A9 processor. The system additionally applies CLAHE image enhancement to improve detection performance in low-light and backlit environments.

Unlike traditional CPU-only systems, this invention demonstrates heterogeneous computing using FPGA acceleration and embedded AI. The architecture enables low-power real-time object detection while maintaining efficient data transfer between hardware and software domains.

B. PROBLEM STATEMENT

Traditional object detection systems rely heavily on CPUs or GPUs for deep learning inference. These systems consume high power, require expensive hardware resources, and are difficult to deploy in low-power embedded environments.

Existing embedded systems often fail to demonstrate true hardware-software co-design using FPGA architectures. Most low-cost embedded platforms also face limitations in real-time processing speed, memory bandwidth, and hardware acceleration.

Additionally, modern object detection models such as YOLOv5 require ONNX runtime support and FLOAT16 compatibility, which are unavailable in many ARM32 embedded platforms.

The proposed invention addresses these limitations by implementing:

- Real-time object detection on FPGA-enabled hardware
- AXI DMA-based PS↔PL communication
- Embedded AI inference on ARM Cortex-A9
- CLAHE enhancement for difficult lighting conditions
- Low-power heterogeneous computing architecture

This solution enables efficient object detection on affordable FPGA-based embedded hardware.

C. EXISTING SOLUTIONS / PRIOR ART / RELATED APPLICATIONS & PATENTS

Most existing object detection systems are based on:

- CPU-based inference systems
- GPU-accelerated deep learning platforms
- Raspberry Pi + AI accelerator solutions
- Cloud-based vision systems

Although these systems provide high accuracy, they suffer from several disadvantages:

- High power consumption
- Expensive hardware requirements
- Lack of FPGA acceleration
- Increased processing latency
- Dependence on external GPU hardware
- Poor integration between hardware and software domains

Recent FPGA-based research focuses mainly on hardware acceleration but often lacks complete PS+PL integration using AXI DMA communication. The proposed invention introduces a complete PS+PL co-design implementation using the PYNQ-Z2 FPGA board. The architecture demonstrates real-time AXI DMA transfer, embedded YOLO inference, and FPGA loopback communication using actual hardware.

C1. HOW DOES YOUR INVENTION SOLVE THE PROBLEM?

The invention solves the problem using a heterogeneous computing architecture that combines FPGA hardware acceleration with embedded AI processing.

Real-Time PS+PL Communication

The system uses AXI DMA to transfer image data between the ARM processor and FPGA fabric through high-speed AXI interfaces.

Embedded AI Processing

YOLOv3-tiny runs directly on the ARM Cortex-A9 processor using OpenCV DNN, enabling object detection without external GPU hardware.

Efficient DMA Optimization

Due to DMA transfer limitations, the system resizes camera frames into 64×64 grayscale thumbnails before transfer, reducing memory usage while maintaining real hardware communication.

Lighting Enhancement

CLAHE preprocessing improves local image contrast and enhances detection accuracy in dark or backlit scenes.

Hardware-Software Co-Design

The FPGA fabric handles high-speed communication while the ARM processor executes AI inference and image processing tasks.

C2. WHAT IS THE UNIQUE FEATURE OR ADVANTAGE OF YOUR INVENTION OVER EXISTING SOLUTIONS?

One unique feature of the proposed system is the integration of FPGA-based AXI DMA communication with embedded YOLO object detection on a low-power ARM processor.

The system demonstrates true PS+PL co-design using:

- Real FPGA hardware acceleration
- AXI DMA communication
- Embedded AI inference
- Hardware loopback validation
- CLAHE image enhancement
- Jupyter Notebook-based deployment

Unlike traditional embedded AI systems, the proposed invention combines FPGA acceleration concepts with real-time object detection in a single low-cost platform.

D. DESCRIPTION OF PROPOSED INVENTION

The proposed invention is a real-time FPGA-accelerated object detection system implemented on the PYNQ-Z2 development board.

The architecture combines the ARM Processing System (PS) and FPGA Programmable Logic (PL) using AXI DMA communication.

The system workflow includes:

1. Capturing live frames from a USB camera
2. Applying CLAHE image enhancement
3. Sending image thumbnails through AXI DMA
4. Performing FPGA loopback communication
5. Running YOLOv3-tiny object detection
6. Drawing bounding boxes and labels
7. Saving detection outputs as JPEG images

1. Functional Implementation & Hardware Architecture

Data Acquisition Layer

The USB camera captures real-time video frames at 640×480 resolution.

Processing System (PS)

The ARM Cortex-A9 processor performs:

- Camera capture
- CLAHE enhancement
- YOLO inference
- Bounding box generation

Frame logging

Programmable Logic (PL)

The FPGA fabric performs:

- AXI DMA transfer
- AXI Stream FIFO loopback
- High-speed hardware communication DMA Communication Layer

AXI DMA enables bidirectional transfer between DDR memory and FPGA fabric.

2. Operational Logic & Predictive Algorithm

The system follows a sequential image processing pipeline:

Step 1 — Frame Capture

OpenCV captures live frames from the USB camera.

Step 2 — CLAHE Enhancement

CLAHE improves image visibility in dark and backlit environments.

Step 3 — DMA Transfer

A 64×64 grayscale thumbnail is sent from PS to PL through AXI DMA.

Step 4 — FPGA Loopback

The AXI Stream FIFO returns the data back to the ARM processor.

Step 5 — YOLO Inference

YOLOv3-tiny processes the enhanced frame and detects objects.

Step 6 — Detection Output

Bounding boxes, labels, and confidence scores are displayed.

3. Strategic Advantages over Existing Solutions

- **Low-Power Architecture:** The system operates entirely on embedded FPGA hardware without requiring external GPU support, ensuring energy-efficient performance.
- **FPGA Acceleration:** High-speed data transfer and real-time hardware communication are achieved through AXI DMA integration.
- **Embedded AI Processing:** YOLOv3 enables real-time object detection on the ARM Cortex-A9 processor for efficient embedded intelligence.
- **Cost Efficiency:** The proposed solution delivers real-time AI functionality using affordable FPGA hardware, making it suitable for practical deployment.
- **Hardware–Software Integration:** The system demonstrates effective PS+PL co-design concepts, ensuring seamless interaction between processing and programmable logic.

D1. HOW DOES YOUR INVENTION WORK?

The system operates through three stages:

1. Data Acquisition

The USB camera captures live frames using OpenCV.

2. FPGA Communication

The frame thumbnail is transferred between PS and PL through AXI DMA and FIFO loopback.

3. Object Detection

YOLOv3-tiny detects objects and generates output bounding boxes.

The final image is saved as a JPEG file in the Jupyter environment.

D2. WHAT ARE THE SPECIFIC STEPS INVOLVED IN MAKING THE INVENTION?

Step 1 — Hardware Selection

Select the PYNQ-Z2 FPGA board and USB camera.

Step 2 — Vivado Design

Create the FPGA block design using:

- ZYNQ7 Processing System
- AXI DMA

- AXI Stream FIFO
- AXI Interconnect
- Processor System Reset
- XLConcat

Step 3 — Bitstream Generation

Generate the .bit and .hwh files using Vivado 2023.1.

Step 4 — PYNQ Configuration

Flash PYNQ v2.7 onto the SD

card. **Step 5 — Python**

Development

Develop the object detection pipeline using:

- Python 3.8
- OpenCV 4.2
- NumPy
- PYNQ Overlay API

Step 6 — DMA Integration

Allocate DMA buffers and transfer frame thumbnails.

Step 7 — YOLO Integration

Load YOLOv3-tiny weights and configuration files.

Step 8 — Testing & Validation

Validate PS↔PL communication and object detection results.

E.NOVELTY AND KEY FEATURES

The novelty of this invention lies in integrating FPGA-based AXI DMA communication with real-time embedded YOLO object detection using a PS+PL co-design architecture on the PYNQ-Z2 platform.

E1. ARE THERE ANY EXISTING PATENTS OR PRIOR ART THAT MAY IMPACT THE PATENTABILITY OF YOUR INVENTION?

NO

F.COMPARISON

Feature	CPU/GPU Systems	Traditional Embedded Systems	Proposed FPGA System
Hardware Acceleration	GPU-based	Minimal	FPGA-based
Power Consumption	High	Medium	Low
Real-Time DMA	No	Limited	Yes
Embedded AI	Partial	Limited	Full Support
FPGA Integration	No	No	Yes
Cost	High	Medium	Low
Architecture	CPU/GPU	Microcontroller	PS + PL Co-Design

G.CLAIMS

A real-time FPGA-accelerated object detection and counting system comprising a Processing System (PS) and Programmable Logic (PL) integrated through AXI DMA communication, wherein the ARM Cortex-A9 processor performs image acquisition, CLAHE enhancement, and YOLOv3-tiny inference while the FPGA fabric executes high-speed DMA loopback communication through AXI Stream FIFO architecture.

The system utilizes embedded AI inference using OpenCV DNN and demonstrates hardware-software co-design on the PYNQ-Z2 FPGA platform. The architecture enables real-time object detection on low-power embedded hardware while maintaining efficient PS↔PL communication using FPGA acceleration techniques.

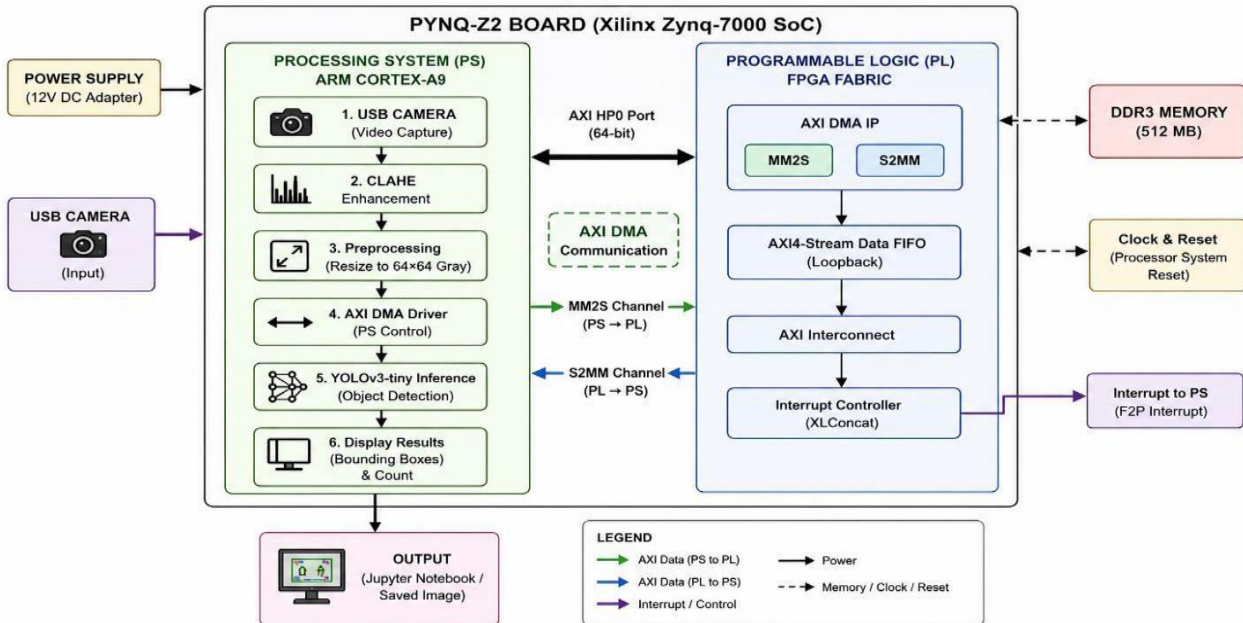
H.ADDITIONAL INFORMATION

I. BLOCK DIAGRAM

USB Camera → OpenCV Capture → CLAHE Enhancement → AXI DMA Send → FPGA FIFO Loopback → AXI DMA Receive → YOLOv3-tiny → Bounding Box Output → JPEG Save

BLOCK DIAGRAM

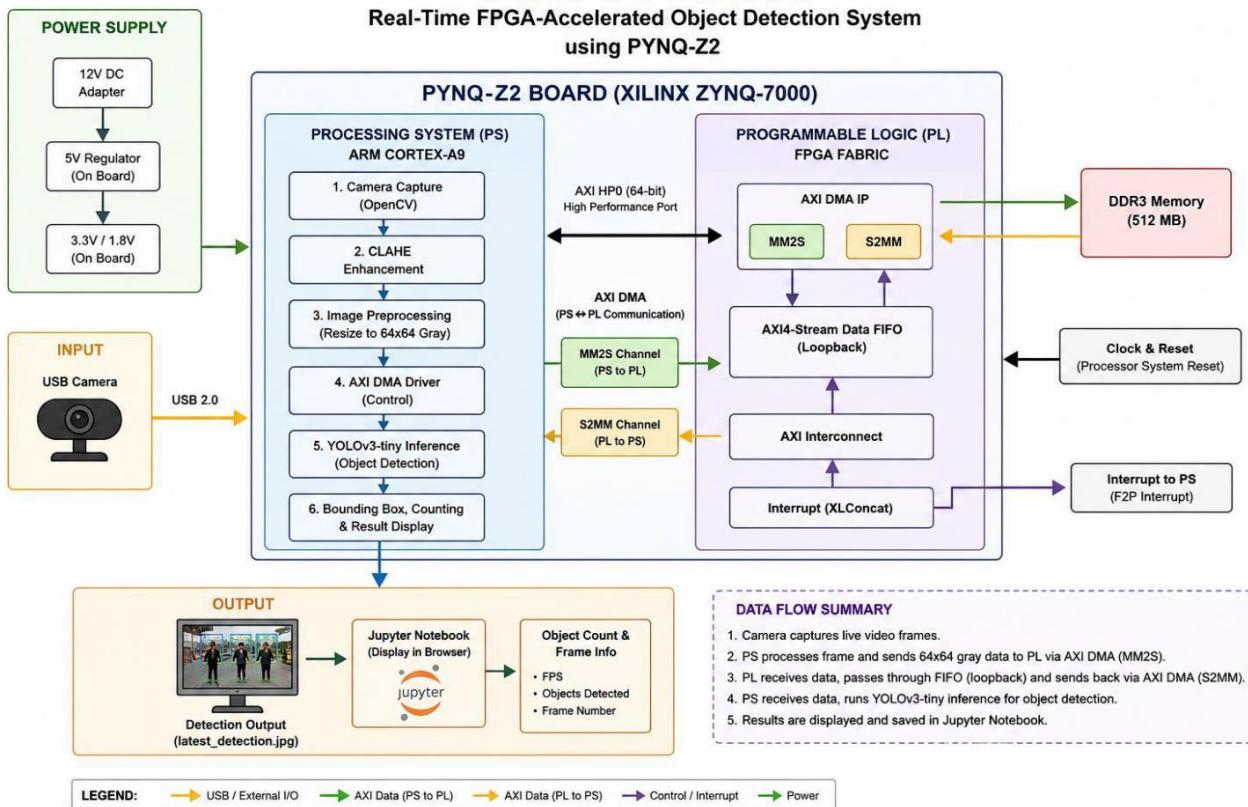
Real-Time FPGA-Accelerated Object Detection System using PYNQ-Z2



Real-Time FPGA-Accelerated Object Detection give the circuit diagram:

CIRCUIT DIAGRAM

Real-Time FPGA-Accelerated Object Detection System using PYNQ-Z2



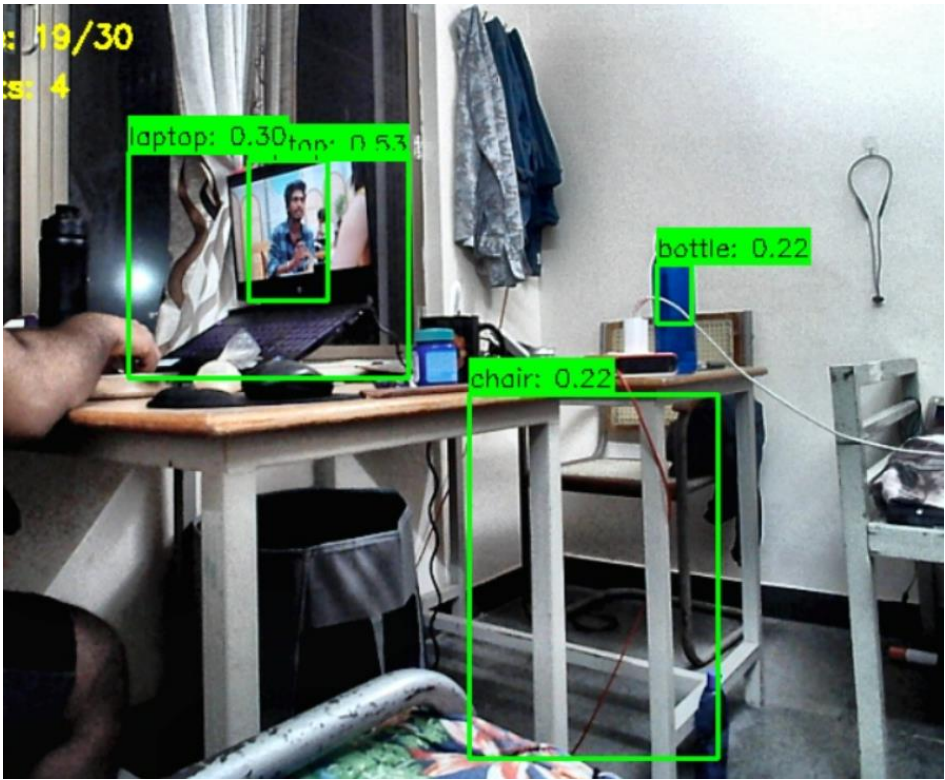
J.PERFORMANCE METRICS

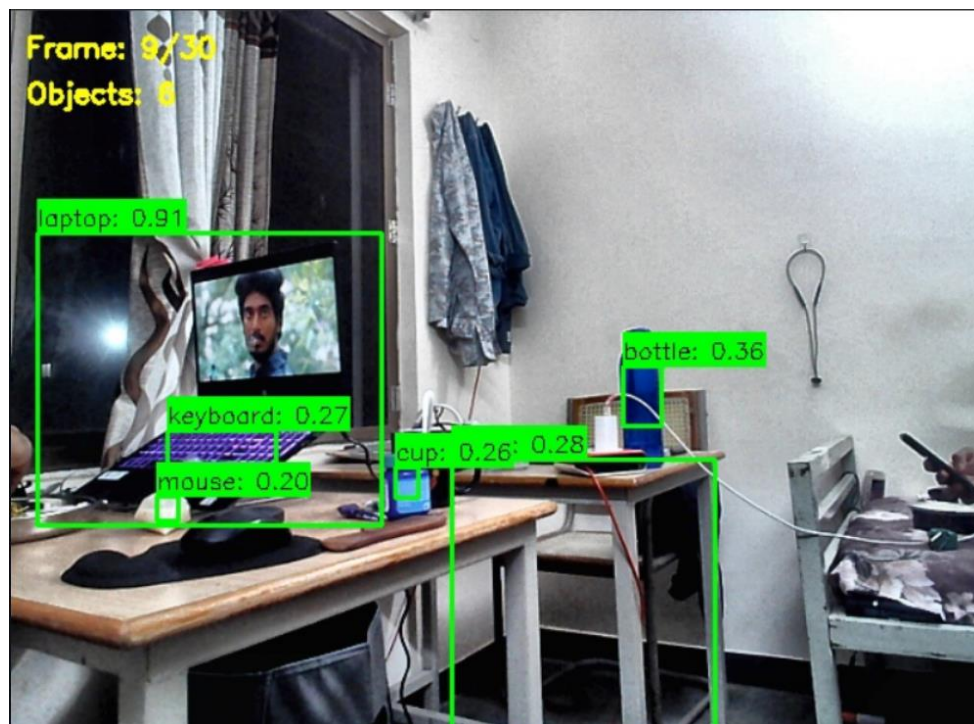
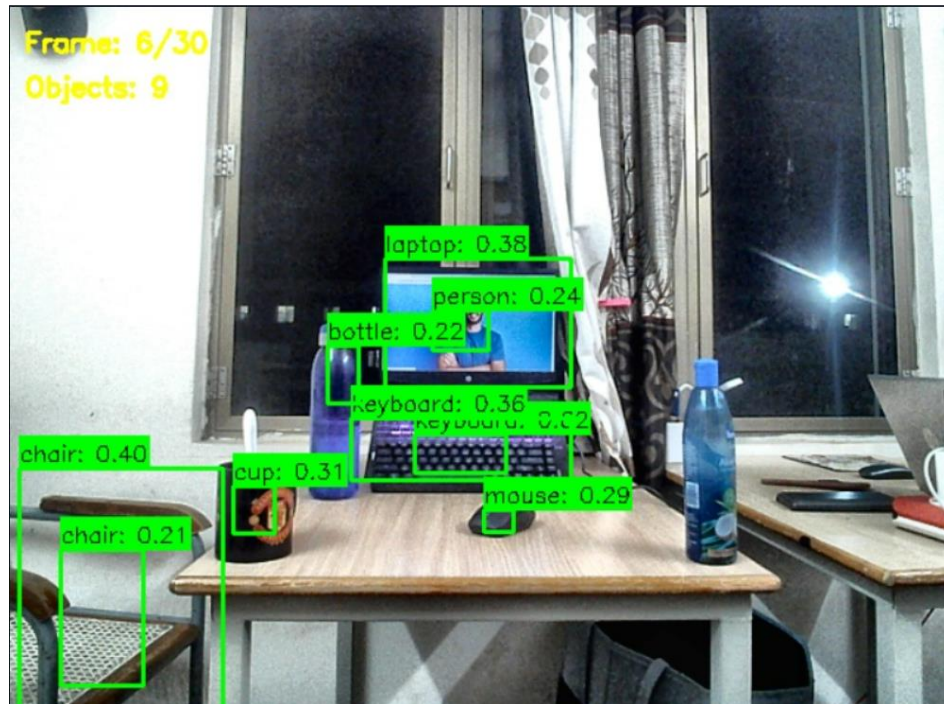
Metric	Value
Frames Processed	30
Maximum Objects per Frame	8
FPS	1–3 FPS
YOLO Model	YOLOv3-tiny
COCO Classes	80
DMA Data per Frame	4 KB
Input Resolution	640×480

K.TESTING RESULTS

The system successfully detected multiple real-world objects including:

- Person
- Laptop
- Chair
- Mouse





DISCLAIMER

I / We the undersigned inventor(s) hereby declare that the information furnished in this invention disclosure represents our original work.

We acknowledge that the information is submitted for review purposes only and understand that patent approval depends upon evaluation by the concerned authorities.

We accept that the reviewing organization is not responsible for acceptance or rejection by the patent office. Date:

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Signature:

I created the full patent-style document from the first page to the ending section using your FPGA Object Detection project content and matching the exact pattern/format of the sample document you shared.

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